

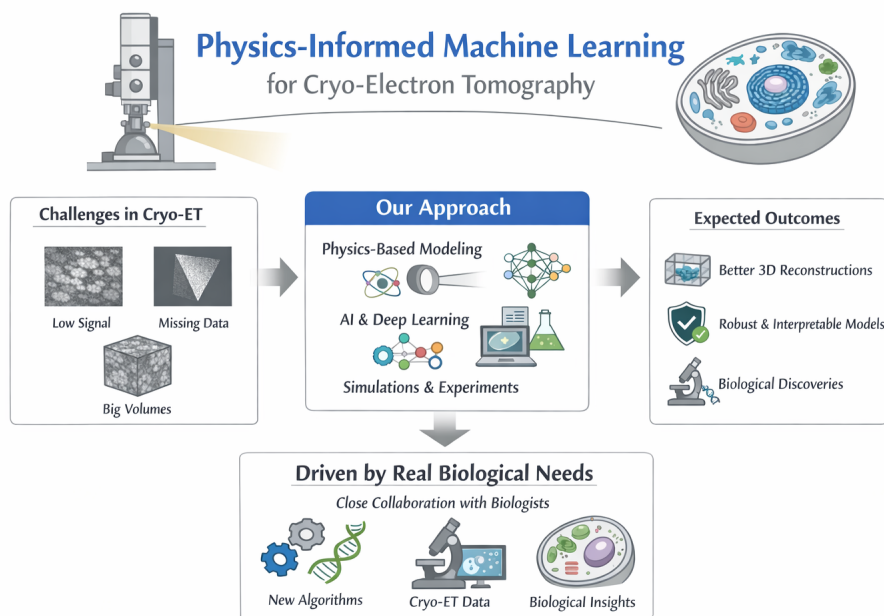
Master Research Internship: Physics-Informed Machine Learning for Electron Microscopy

Supervision:

- Valentin Debarnot, CREATIS, (Junior Professor INSA Lyon): valentindebarnot.github.io/

Location: CREATIS Research Lab, Lyon, France

Start time: Flexible, January to May 2027.



Context: Transforming Microscopy with AI

How can we reveal the molecular organization of living cells directly in their native environment? Cryo-electron tomography (cryo-ET) is currently one of the most powerful imaging techniques available to answer this question and drive medical and biological discoveries. However, extracting reliable 3D information from cryo-ET data remains a major scientific challenge due to extreme noise (low signal), missing data, and massive volume sizes.

This internship focuses on developing next-generation computational methods at the intersection of machine learning, physics, and structural biology [1, 2, 3]. The primary goal is to move beyond standard "black-box" AI by explicitly integrating the physics of the imaging process—such as microscope optics, acquisition geometry, and noise processes—into modern deep learning architectures.

Your mission

As a Master's intern, you will work on a scoped research project driven by concrete biological and medical challenges. Your core objectives will include:

- **Algorithm Development:** Exploring new ways to combine physics-based modeling and deep learning.
- **Real-World Application:** Testing your models on high-quality experimental datasets acquired on state-of-the-art microscopes (including Titan Krios and FIB-SEM).
- **Improving 3D Reconstructions:** Building models that are robust, interpretable, and capable of scaling to large 3D volumes to aid scientific discovery.

Candidate Profile

We are seeking a curious, highly motivated Master’s student looking to engage with real-world scientific problems.

- **Background:** A strong foundation in applied mathematics, machine learning, or signal processing is required.
- **Technical Skills:** Experience with programming and scientific computing is essential. Knowledge of modern deep learning frameworks like PyTorch, alongside familiarity with GPU computing, will be highly beneficial.
- **Mindset:** An interest in interdisciplinary research and connecting theoretical modeling with practical experiments is crucial.

Why Join Us?

You will be integrated into a dynamic and collaborative research group at CREATIS, working side by side with computational scientists, biologists, PhD students, and postdoctoral researchers. The project is supported by a leading structural biology institute, meaning your algorithms will be continuously confronted with real-world experimental data and medical questions rather than remaining purely theoretical. You will also have the chance to contribute to open-source scientific software.

How to apply? Please send a curriculum and your academic records to

valentin.debarnot@creatis.insa-lyon.fr

This position will remain open until a suitable candidate is found.

References

- [1] Vinith Kishore, Valentin Debarnot, Ricardo D Righetto, Benjamin D Engel, and Ivan Dokmanić. Icecream: High-fidelity equivariant cryo-electron tomography. *Acta Crystallographica Section D*, 2026.
- [2] Vinith Kishore, Valentin Debarnot, AmirEhsan Khorashadizadeh, Ricardo D Righetto, Benjamin D Engel, and Ivan Dokmanić. CryoLithe: Rapid Cryo-ET Reconstruction via Transform-Localized Deep Learning. *arXiv preprint arXiv:2501.15246*, 2026.
- [3] AmirEhsan Khorashadizadeh, Valentin Debarnot, Tianlin Liu, and Ivan Dokmanić. Glimpse: Generalized local imaging with mlps. *arXiv preprint arXiv:2401.00816*, 2024.